

Joshua Dongmin Keum

KEUM, DONGMIN

EDUCATION

Seoul National University, Seoul, Republic of Korea

B.Sc., Chemistry — *Minor*, Statistics

August '28 (Expected)

GPA: 4.07/4.3 overall; 4.15/4.3 in Chemistry

Leave of Absence for Mandatory Military Service: Oct. '25 - Jun. '27

RESEARCH EXPERIENCE

Laboratory of Biomolecular Diversity

Supervisor : Prof. Seokhee Kim

Undergraduate Research Assistant

June '24 - July '25

I. Biochemical investigation of sequential peptide oxidative decarboxylation and cross-linkage

- Constructed and optimized precursor peptide expression plasmid with MBP and TEV cleavage tags
- Expressed and purified recombinant proteins, and assayed LanD-mediated C-terminal AviCys formation and subsequent P450-mediated cross-linkage by negative-mode MALDI-TOF MS

II. Development of a HMM-based prediction filter of RiPP biosynthetic products

- Constructed and validated custom HMM profiles for RiPP biosynthetic enzymes across 20+ classes, including lasso, rSAM-mediated, and quorum sensing peptides.
- Established E-value thresholds for product prediction using statistical analysis of curated databases
- Developed a Python tool to predict and visualize domain architectures in pangenomic contigs using Pfam and custom HMMs
- Framework subsequently extended to 59 classes by a colleague, achieving 93% precision and 70% recall across 2,636 RiPP-annotated entries of MIBiG database
- Code: Known_RiPP_filter ([GitHub](#))

III. Genome mining of response regulators associated with secondary metabolite biosynthesis

- Identified 3,297 homologous response regulators via PSI-BLAST genome mining
- Mapped biosynthetic gene clusters in genomic neighborhoods surrounding the response regulators, and identified 549 bufferin-like clusters
- Discovered a novel YcaO-homolog with unique N- and C-terminal domains, suggesting a distinct peptide activation mechanism

IV. Bioinformatic expansion of Pgm1-like ATP-grasp enzymes in natural product biosynthesis

- Identified 1,923 ATP-grasp ligases by PSI-BLAST genome mining from the bacterial pangenome
- Recovered known Pgm1-associated gene clusters, corroborating the mining strategy
- Identified novel gene clusters predicted to catalyze amide bond formation between diverse natural product classes, such as type III polyketides, NRPs, RiPPs, polysaccharides, and L-propargylglycine.

Continuing Research Collaborator (Remote)

May '26 - Present

- Identified a previously uncharacterized radical SAM enzyme acting on precursor peptides with C-terminal Thr precursors through systematic search of candidate RiPP BGCs. Genome mining reveals co-occurrence with the backbone amidine-installing enzyme ItfB-like enzymes in some genomic contexts.
- Continued development of the RiPP annotation framework during mandatory military service

PRESENTATIONS	Poster Presentation, Seoul National University, November 2024, Keum, D.; Park, Y.; Kim, S. “Bioinformatic Expansion on Response Regulators for Novel Natural Product Discovery.”
HONORS & AWARDS	Institutional Merit-based Scholarship, Seoul National University <i>Spring '23 – Spring '25</i> Commendation for Exemplary Student 2024, College of Natural Sciences, Seoul National University Dean’s List, College of Natural Sciences, Seoul National University <i>Fall '22, Fall '23 – Spring '25</i>
TECHNICAL SKILLS	Computational & Theoretical • Programming: Python, $\text{\LaTeX}$, R • Bioinformatics: Biopython, BLAST/PSI-BLAST, antiSMASH, HMMER; HMM construction, MSA, phylogenetics, pangenomic BGC analysis • Structural Biology: PyMOL, AlphaFold, Foldseek • Statistical Analysis: Regression analysis, statistical design of experiments, and statistical classification and clustering. Wet Laboratory • Molecular Biology: PCR, cloning, and plasmid transformation; Heterologous expression in <i>E. coli</i>, Ni-NTA histidine tag affinity chromatography, HPLC, SDS-PAGE, MALDI-TOF MS
TEACHING & LEADERSHIP	President, 17th Student Council, Department of Chemistry, Seoul National University <i>November '23 - October '24</i> • Organized and coordinated 5+ department-wide events involving undergraduates, graduates, and faculty. • Initiated and led the department’s first Curriculum Improvement Committee to incorporate undergraduate feedback into curriculum discussions. Peer Tutor, <i>Basic Calculus I (F31.101)</i>, Seoul National University <i>Spring '23, Fall '23, Spring '25</i>
OTHER ACTIVITIES	Member, RxN Journal Club (Undergraduate Chemistry Research Society) Department of Chemistry, Seoul National University <i>Spring '24, Spring '25</i> • Biocatalysis (Spring '24): 4 presentation sessions with topics including engineering catalytic function into non-enzymatic proteins. • Biochemistry — Structure–Function Relationships (Spring '25): 3 presentation sessions with topics including biosynthetic gene clusters and lncRNA-mediated regulation of phenylalanine hydroxylase.